

Erdős Problem 776: exact thresholds for antichains with level multiplicity

Human-led, AI-assisted research package

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Status and principal statement

For the threshold $n_0(r)$ defined for $r \geq 2$, the combined result is

$$n_0(r) = \begin{cases} 3, & r = 2, \\ 8, & r = 3, \\ 2r + 4, & 4 \leq r \leq 10, \\ 2r + 5, & r \geq 11. \end{cases}$$

The values at $r=2,3$ are due to Yixin He and Quanyu Tang. Lean proves the original least-threshold statement for every $r \geq 4$: $4 \leq r \leq 10$ uses one closed seven-value certificate, $11 \leq r \leq 28$ one closed eighteen-value certificate, $29 \leq r \leq 377$ one ground-bounded lower-window certificate, and $r \geq 378$ a symbolic proof. The checker definitions and soundness bridges are ordinary Lean definitions and proofs; the three closed finite evaluations use `native_decide` and therefore additionally trust Lean’s native evaluator/compiler. Four separate adversarial AI review passes attempted to falsify the argument. They are not treated as independent specialist peer review.

The load-bearing proof is split into the following four parts in this PDF:

1. the prescribed-profile criterion and exact $r=11$ result;
2. the finite exact range $4 \leq r \leq 10$;
3. the L4 reverse-envelope theorem;
4. the uniform $r \geq 11$ construction and obstruction.

The source repository contains the original-formulation Lean statement, exact certificates, standard-library verification scripts, theorem crosswalk, axiom audit, proof guide, audit trail, and contribution record.

Part I: prescribed profiles and the exact case $r=11$

An exact calculation at $r = 11$ in the Erdős-Trotter antichain problem

Revised note, 17 July 2026. The result below is included in the end-to-end Lean theorem for every $r \geq 4$; independent specialist peer review remains pending.

1. Definitions and result

For a family $\mathcal{F} \subseteq 2^{[n]}$, let

$$S(\mathcal{F}) = \{|A| : A \in \mathcal{F}\}.$$

An r -**multiplicity antichain** is an antichain for which every occurring level contains at least r members. Let $g(n, r)$ be the maximum possible value of $|S(\mathcal{F})|$. Following He and Tang, let $n_0(r)$ be the least threshold such that $g(n, r) = n - 3$ for every $n > n_0(r)$.

The result proved here is:

Theorem.

$$g(27, 11) = 23 \quad \text{and} \quad n_0(11) = 27.$$

For completeness, we first record the elementary upper bound needed throughout the proof. Let $n \geq 12$. In an 11-multiplicity antichain, levels 0 and n cannot occur, since each contains only one set. If level 1 occurs, choose 11 singleton members. Every other member must avoid those 11 points, so the occupied sizes lie in $\{1\} \cup \{2, 3, \dots, n-11\}$; hence at most $n-11 < n-3$ levels occur. Applying the same argument to the complement family handles level $n-1$. Otherwise all occupied levels lie among $2, 3, \dots, n-2$, of which there are $n-3$. Thus

$$g(n, 11) \leq n - 3,$$

and equality forces the occupied levels to be exactly $2, 3, \dots, n-2$. In particular, a hypothetical 24-level witness at $(n, r) = (27, 11)$ must occupy exactly $2, 3, \dots, 25$, with at least 11 members on each level. The lower-bound necessity theorem therefore obstructs such a witness directly, while recurrence feasibility constructs one with exactly 11 members on each level. Consequently,

$$g(27, 11) = 24 \iff \text{the profile } f_i = 11 \text{ } (2 \leq i \leq 25) \text{ is feasible.}$$

2. Exact profile criterion

For $m \geq 0$ and $k \geq 1$, write the unique binomial (cascade) representation

$$m = \binom{a_k}{k} + \binom{a_{k-1}}{k-1} + \dots + \binom{a_t}{t}, \quad a_k > a_{k-1} > \dots > a_t \geq t,$$

and define

$$\partial_k(m) = \binom{a_k}{k-1} + \binom{a_{k-1}}{k-2} + \dots + \binom{a_t}{t-1}, \quad \partial_k(0) = 0.$$

Kruskal-Katona says that $\partial_k(m)$ is the minimum possible lower shadow of m distinct k -sets, attained by a colex initial segment.

Let $f = (f_0, \dots, f_n)$ be a proposed profile and let $s = \max\{i : f_i > 0\}$. Define

$$m_s = f_s, \quad m_i = f_i + \partial_{i+1}(m_{i+1}) \quad (i = s-1, s-2, \dots, 0).$$

The form used in this package is deliberately asymmetric.

Necessity from lower bounds. If an antichain has at least f_i members on every level i , then

$$m_i \leq \binom{n}{i}$$

for every i .

Sufficiency with equality. Conversely, if all these capacities fit, the colex construction produces an antichain with exactly f_i members on every level i .

In particular, for an exact proposed profile this gives the usual if-and-only-if criterion. The asymmetric statement avoids any trimming step in obstruction arguments.

This is a formulation of the classical squashed-antichain profile theorem. A short proof is included so that the profile reduction and greedy construction can be checked locally; Kruskal-Katona itself, in the cascade form stated above with colex attainment, is used as the classical black box.

Necessity

For an antichain $\mathcal{F}$, let $\mathcal{F}_i$ be its level- i part. Let D_i be the collection of all i -sets properly contained in some member of $\mathcal{F}$ on a higher level, and set

$$M_i = \mathcal{F}_i \dot{\cup} D_i.$$

The union is disjoint because $\mathcal{F}$ is an antichain. Moreover, the lower shadow of M_{i+1} lies inside D_i . Therefore

$$|M_i| \geq f_i + \partial_{i+1}(|M_{i+1}|).$$

At the highest requested level, whether or not there are family members above it, M_s contains the level- s slice of the antichain, so $|M_s| \geq f_s = m_s$. Starting from this base and using monotonicity of the Kruskal-Katona function gives $|M_i| \geq m_i$ by downward induction. Since M_i consists of i -subsets of $[n]$, this implies $m_i \leq \binom{n}{i}$.

Sufficiency

Choose the family from the top down. At the highest occupied level take the first f_s sets in colex. Inductively, suppose that at level $i+1$ the union of the down-closure of all higher chosen sets with the sets chosen on level $i+1$ is the colex initial segment of size m_{i+1} . Its

lower shadow is therefore the colex initial segment of size $\partial_{i+1}(m_{i+1})$. Moreover, an i -set lies below some chosen set on a level above i if and only if it lies below an intermediate $(i+1)$ -set in this cumulative segment; this is just transitivity of containment through level $i+1$. Hence that shadow is exactly the level- i down-closure of all higher choices.

Choose the next f_i sets in colex, immediately following this shadow. The hypothesis $m_i \leq \binom{n}{i}$ ensures that these sets lie in $[n]$, and their union with the shadow is the colex initial segment of size m_i . This preserves the induction invariant. The newly chosen lower sets avoid the down-closure of all higher chosen sets, so the resulting family is an antichain with profile f .

3. The calculation at $(n, r) = (27, 11)$

For the 24-level profile

$$f_i = 11 \quad (2 \leq i \leq 25),$$

the recurrence gives

$$m_3 = 2709 = \binom{26}{3} + \binom{15}{2} + \binom{4}{1}.$$

Therefore

$$\partial_3(m_3) = \binom{26}{2} + \binom{15}{1} + \binom{4}{0} = 341, \quad m_2 = 341 + 11 = 352 > 351 = \binom{27}{2}.$$

The level-2 failure already proves infeasibility. For accuracy, level 1 also fails: the 2-cascade of $352 = \binom{27}{2} + \binom{1}{1}$ gives $m_1 = 27 + 1 = 28 > 27$. Thus no 11-multiplicity antichain on $[27]$ can use 24 distinct sizes, and

$$g(27, 11) \leq 23.$$

For the profile

$$f_i = 11 \quad (2 \leq i \leq 24),$$

the recurrence fits at every level. The useful bottom calculation is

$$m_3 = 2413 = \binom{25}{3} + \binom{15}{2} + \binom{8}{1}, \quad \partial_3(m_3) = 316, \quad m_2 = 327 \leq 351.$$

The actual tight capacity occurs one level lower: $327 = \binom{26}{2} + \binom{2}{1}$, so $m_1 = \partial_2(327) = 26 + 1 = 27 = \binom{27}{1}$.

The colex construction therefore supplies 11 sets on each of the 23 levels $2, \dots, 24$. The generated family has 253 sets and passes a separately implemented pairwise-containment checker. Hence

$$g(27, 11) = 23.$$

The complete arithmetic and explicit construction are reproduced by `tests/verify_r11.py`; the explicit family is in `certificates/r11_n27_23_levels.txt`.

4. Construction for every $n \geq 28$

We use two elementary lemmas.

Core-to-full lemma

Suppose $\mathcal{C}$ is an antichain on $[q]$ having at least r sets of every size

$$2, 3, \dots, \lfloor n/2 \rfloor - 1,$$

where $q \leq n - 1 - r$. Introduce a pivot point a and r private points $p_1, \dots, p_r$, all outside the core. Form the lower family from

$$\{a, p_j\} \quad (1 \leq j \leq r)$$

and, for each required core level, r sets $C \cup \{a\}$. Add their complements to supply the upper levels (at the middle level when n is even, retaining either side already supplies at least r sets).

The lower family is an antichain. Distinct pairs have the same size; pivoted core sets satisfy $C \cup \{a\} \subseteq C' \cup \{a\}$ exactly when $C \subseteq C'$, which the core antichain excludes; and the private points prevent a pair $\{a, p_j\}$ from lying inside a pivoted core set. Taking complements reverses containment, so the complemented upper family is also an antichain.

It remains to check both cross-side directions. If a lower set L were contained in the complement of another lower set L' , then L and L' would be disjoint, although both contain a . Conversely, $[n] \setminus L' \subseteq L$ would imply $L \cup L' = [n]$. This is impossible because both lower sets contain a and have size at most $\lfloor n/2 \rfloor$, giving

$$|L \cup L'| \leq 2\lfloor n/2 \rfloor - 1 \leq n - 1.$$

Thus the result is an r -multiplicity antichain using every size $2, \dots, n - 2$.

Two-point padding lemma

Let $\mathcal{C}$ be an antichain on $[q]$ with at least r sets of every size $2, \dots, s$. Suppose that $D_1, \dots, D_r$ are distinct $(s - 1)$ -subsets of $[q]$, none of which contains a member of $\mathcal{C}$. With two new points x, y , set

$$\mathcal{C}' = \mathcal{C} \cup \{D_i \cup \{x, y\} : 1 \leq i \leq r\}.$$

The new sets are mutually incomparable. If an old set were contained in a new one, it would be contained in D_i , contrary to the hypothesis; a new set cannot be contained in an old one because it uses new points. Therefore $\mathcal{C}'$ is an antichain and supplies level $s + 1$.

Freeness persists indefinitely in a completely explicit way. Fix base free sets $G_1, \dots, G_r$. If the first t added pairs are $\{x_j, y_j\}$, define

$$D_i^{(t)} = G_i \cup \{x_1, \dots, x_t\}.$$

This set contains no base-core member. It also cannot contain a member born at stage j , because every such member contains y_j , whereas $D_i^{(t)}$ omits y_j . Hence the next padding step can add

$$D_i^{(t)} \cup \{x_{t+1}, y_{t+1}\} \quad (1 \leq i \leq r),$$

and the same fixed base free sets sustain the chain indefinitely.

The $r = 11$ base

The exact profile criterion constructs a core on 16 points with 11 sets of every size $2, \dots, 13$. Direct enumeration finds 88 free 12-subsets; only 11 are required. Both the antichain property and the free-set count are checked in `tests/verify_r11.py`. The complete 132-set core and the first 11 free 12-subsets are explicitly listed in `certificates/r11_core_and_free_sets.txt`, which the verifier parses and checks without regenerating the core.

At padding stage t , the core has

$$q = 16 + 2t, \quad s = 13 + t.$$

For $n = 28 + 2t$, these are exactly the parameters needed by the core-to-full lemma; for $n = 29 + 2t$, the same core fits with one point to spare. Hence a full-profile 11-multiplicity antichain exists for every $n \geq 28$.

Since the full profile fails at $n = 27$, this proves

$$n_0(11) = 27.$$

5. Relation to the uniform companion result

This note deliberately keeps the exact $r = 11$ proof self-contained. A separate companion manuscript in [UNIFORM_THEOREM.md](#) gives an AI-assisted proof, now formalized in Lean, for

$$n_0(r) = 2r + 5 \quad (r \geq 11).$$

The exact calculation and stored certificates in this note do not depend on the uniform argument. Conversely, the uniform manuscript uses the same classical profile criterion and the core/padding framework developed here. The uniform argument has passed four adversarial AI audits but has not yet undergone independent specialist peer review; the repository records that distinction explicitly.

References

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4. D. E. Daykin, J. Godfrey, and A. J. W. Hilton, *Existence Theorems for Sperner Families*, *Journal of Combinatorial Theory, Series A* **17** (1974), 245–251.

5. P. Lieby, *Antichains on Three Levels*, Electronic Journal of Combinatorics **11** (2004), R50. Theorem 2.8 records the classical prescribed-profile squashing theorem: <https://www.combinatorics.org/ojs/index.php/eljc/article/view/v11i1r50>.

Part II: exact thresholds for $4 \leq r \leq 10$

Exact thresholds for multiplicities $4 \leq r \leq 10$

This note completes the finite range between the exact values of He-Tang at $r=2,3$ and the uniform theorem for $r \geq 11$.

Throughout, $g(n, r)$ is the maximum number of occupied levels in an r -multiplicity antichain on $[n]$, and $n_0(r)$ is the least integer n_0 such that

$$g(n, r) = n - 3 \quad \text{for every } n > n_0.$$

The definition of $n_0(r)$ is used for $r \geq 2$. The original problem treats $r=1$ separately, where the extremal number of occupied levels is $n-2$ rather than $n-3$.

Theorem

For every integer r with $4 \leq r \leq 10$,

$$n_0(r) = 2r + 4.$$

Together with He-Tang's exact values

$$n_0(2) = 3, \quad n_0(3) = 8,$$

and the companion uniform theorem

$$n_0(r) = 2r + 5 \quad (r \geq 11),$$

this gives the complete threshold table for every value $r \geq 2$:

$$n_0(r) = \begin{cases} 3, & r = 2, \\ 8, & r = 3, \\ 2r + 4, & 4 \leq r \leq 10, \\ 2r + 5, & r \geq 11. \end{cases}$$

The last line is formalized in Lean together with the finite statement proved in this note. Both retain the package-wide status of pending independent specialist peer review; this finite proof is independent of the symbolic large- r phase argument.

Lean formalizes this entire range in `Erdos776/Uniform/SmallRangeCertificate.lean`: one closed seven-value check certifies the obstruction, the first fitting full profile, and the base core; profile sufficiency, puncturing, padding, and global leastness are then symbolic theorems. The endpoint is `isErdosThreshold_of_ge_4_le_10`.

1. Reduction to a prescribed profile

For $r \geq 4$, an r -multiplicity antichain with $n-3$ occupied levels must use exactly the levels

$$2, 3, \dots, n-2.$$

Indeed, levels 0 and n cannot occur. If level 1 occurs, choose r singleton members. Every other member avoids those r points, so all occupied sizes lie in

$$1, 2, \dots, n-r,$$

and there are at most $n-r \leq n-4$ of them. Complementation excludes level $n-1$ in an $n-3$ -level witness.

Consequently, for $r \geq 4$, equality $g(n, r) = n-3$ is equivalent to feasibility of the full prescribed profile

$$f_i = r \quad (2 \leq i \leq n-2), \quad f_i = 0 \quad \text{otherwise.}$$

We use the classical Clements-Daykin-Godfrey-Hilton criterion in the asymmetric form proved in docs/TECHNICAL_NOTE.md: if

$$m_s = f_s, \quad m_i = f_i + \partial_{i+1}(m_{i+1}),$$

then an antichain with at least f_i members forces

$$m_i \leq \binom{n}{i}$$

at every relevant level. Conversely, these inequalities construct a colex antichain with exactly profile f . Thus no trimming is needed for the obstruction, while the same recurrence gives the exact construction.

2. The obstruction at $n=2r+4$

For each $4 \leq r \leq 10$, run the exact recurrence for the full profile at

$$n = 2r + 4.$$

It overflows at level 2 as follows.

r	$n=2r+4$	m_2	$C(n, 2)$	overflow
4	12	67	66	1
5	14	92	91	1
6	16	121	120	1
7	18	155	153	2
8	20	192	190	2
9	22	233	231	2
10	24	278	276	2

Thus

$$g(2r + 4, r) < 2r + 1 = (2r + 4) - 3,$$

so the threshold cannot be smaller than $2r+4$:

$$n_0(r) \geq 2r + 4.$$

Only this single failed value is needed for the threshold lower bound. If $n_0(r) \leq 2r+3$, the definition would require success at $n=2r+4$.

3. The tight full profile at $n=2r+5$

At

$$n = 2r + 5,$$

the same recurrence is feasible for every $4 \leq r \leq 10$, and it is exactly tight at level 2:

r	$n=2r+5$	m_2	$C(n, 2)$
4	13	78	78
5	15	105	105
6	17	136	136
7	19	171	171
8	21	210	210
9	23	253	253
10	25	300	300

The sufficiency direction of the profile theorem therefore gives an explicit colex antichain on $[2r+5]$ with exactly r members on every level $2, \dots, 2r+3$. These seven witnesses are stored in

`certificates/small_r_full_witnesses_r4_to_10.txt`

and are checked by direct pairwise containment tests, independently of the shadow computation. Lean constructs the same semantic existence result from the checked capacities through `profile_sufficiency`; the stored files are not logical inputs to the Lean theorem.

Hence

$$g(2r + 5, r) = 2r + 2 = (2r + 5) - 3.$$

4. A base core and explicit free sets

For each $4 \leq r \leq 10$, put

$$q = r + 5, \quad s = r + 2.$$

The core profile with r sets on every level $2, \dots, s$ is feasible on q points. Its bottom recurrence values are:

r	q	core m_2	C(q,2)	slack
4	9	31	36	5
5	10	41	45	4
6	11	51	55	4
7	12	62	66	4
8	13	74	78	4
9	14	87	91	4
10	15	101	105	4

The canonical colex core has an explicit family of r free $(s-1)=(r+1)$ -sets. Write

$$X = [q - 2], \quad J = \{4, 5, \dots, q - 2\}.$$

The first $r=q-5$ top-level core sets are

$$T_j = X \setminus \{j\}, \quad j \in J.$$

Indeed, the top state r has a cascade of r unit diagonal digits. Its colex realization consists exactly of the $(r+2)$ -subsets of $X = [r+3]$ obtained by deleting one of the r points in $J = \{4, \dots, r+3\}$.

We use one elementary observation.

Puncturing lemma. If C is an antichain, T belongs to C , and D is a proper subset of T , then D contains no member of C .

Indeed, if A belongs to C and A is contained in D , then A is contained in T . Antichainness forces $A=T$, contradicting the fact that T is not contained in its proper subset D .

For each j in J , puncture the displayed top member at the common point 1:

$$D_j = T_j \setminus \{1\} = [q] \setminus \{1, j, q-1, q\}.$$

There are exactly r distinct such sets, each has size $q-4=r+1$, and D_j is a proper subset of the core member T_j . The puncturing lemma says immediately that every D_j is free. No classification of any other core member is required. The complete stored cores and free sets are in

certificates/small_r_cores_and_free_sets_r4_to_10.txt

5. Padding to every $n \geq 2r+6$

The two-point padding lemma from docs/TECHNICAL_NOTE.md applies without any large- r hypothesis. If a core on q points has r sets on levels $2, \dots, s$ and free sets $D_1, \dots, D_r$ of size $s-1$, then adding two new points x, y and the sets

$$D_i \cup \{x, y\} \quad (1 \leq i \leq r)$$

produces a core with the next occupied level.

Freeness persists indefinitely. After earlier padding pairs $\{x_h, y_h\}$, use

$$D_i^{(t)} = D_i \cup \{x_1, \dots, x_t\}.$$

A base-core member cannot lie in $D_i^{\{t\}}$ because it would lie in D_i , and a member born at stage h contains y_h , which $D_i^{\{t\}}$ omits.

Starting from $(q, s) = (r+5, r+2)$, after t stages the core parameters are

$$(q_t, s_t) = (r + 5 + 2t, r + 2 + t).$$

The core-to-full construction then gives the full profile for both

$$n = 2r + 6 + 2t \quad \text{and} \quad n = 2r + 7 + 2t.$$

Therefore

$$g(n, r) = n - 3 \quad \text{for every } n \geq 2r + 6.$$

Together with the direct full-profile construction at $n=2r+5$, this proves

$$n_0(r) \leq 2r + 4.$$

Combining with Section 2 gives the theorem.

6. The transition at $r=11$

The finite thresholds also explain why the uniform formula changes at $r=11$. Let $A=r+4$, and run the upper core recurrence with r sets on levels $2, \dots, r+2$. Write

$$a_3 = \binom{A}{3} + e_r.$$

Exact cascade arithmetic gives

r	4	5	6	7	8	9	10	11
e_r	-7	-4	-3	-2	-2	-2	-2	1

Thus the core residual remains below the binomial shell through $r=10$, but crosses it at $r=11$. This is the same carry that, in the uniform argument, obstructs the full profile at $n=2r+5$. The threshold jump

$$2r + 4 \rightarrow 2r + 5$$

therefore coincides with a sign change in a single combinadic residual. This observation is explanatory rather than a separate proof dependency.

7. Reproducibility

Run

```
python3 verification/verify_small_r_exact.py  
lake build Erdos776.Uniform.SmallRangeCertificate
```

The script is independent of the project profile implementation. It:

- recomputes every obstruction and feasible profile using its own exact Kruskal-Katona cascade;
- parses and directly checks all seven stored full-profile witnesses;
- parses and directly checks all seven stored cores and free-set families;
- directly checks five persistent padding stages and the resulting full witnesses for both parities at every tested stage.

Expected final lines:

PASS: exact finite obstruction, construction, and padding checks passed ($4 \leq r \leq 10$)

The proof for all larger n is the symbolic persistent-padding argument in Section 5, not an inference from a finite scan.

Part III: the L4 reverse-envelope theorem

L4: a proof that $\alpha_r \leq r - 1$ (all $r \geq 29$)

Standalone companion to UNIFORM_THEOREM.md. The proof written here is symbolic for all $r \geq 29$. In the final Lean theorem the range $11 \leq r < 29$ is closed by a direct bounded upper-core/overflow certificate; verification/verify_l4.py remains an independently implemented exact cross-check of L4 on that finite range. This note uses only the classical Kruskal-Katona shadow function and elementary binomial identities.

Cascade notation

For $m \geq 0$ and $k \geq 1$, write the canonical cascade

$$m = \binom{a_k}{k} + \binom{a_{k-1}}{k-1} + \cdots + \binom{a_s}{s}, \quad a_k > a_{k-1} > \cdots > a_s \geq s,$$

and define

$$\partial_k(m) = \binom{a_k}{k-1} + \binom{a_{k-1}}{k-2} + \cdots + \binom{a_s}{s-1}.$$

We use monotonicity of ∂_k , Pascal's identity, and the fact that shadowing a displayed canonical cascade lowers each position by one.

0. Statement

Here α_r is the largest integer y for which the profile with r sets at every level $2, \dots, r$ and y sets at level $r+1$ is feasible on $A = r+4$ points. Equivalently, it is the largest top value whose canonical profile recurrence fits every capacity; this equivalence is the prescribed-profile criterion.

Theorem L4. For every $r \geq 29$ the profile [r sets of every size $2..r$, r sets of size $r+1$] on $A = r+4$ points is infeasible; equivalently $\alpha_r \leq r-1$. Lean kernel-checks this semantic theorem in Erdos776/L4/Profile.lean. The finite range $11 \leq r < 29$ is checked exactly by verification/verify_l4.py; the final Lean threshold theorem handles that range through its smaller direct certificate rather than importing the Python result.

By DGH/Clements necessity, feasibility would give the chain

$$c_{\{r+1\}} = r, \quad c_i = r + \partial_{\{i+1\}}(c_{\{i+1\}}) \quad (i = r, \dots, 2),$$

with $c_2 \leq C(A, 2)$. We derive a contradiction from $c_2 \leq C(A, 2)$ alone (no other capacity bound is used).

1. The bounding sequence $\hat{G}$

Define, for $r \geq 29$:

$$\hat{G}_2 = C(r+4, 2)$$

$$\hat{G}_3 = C(r+3, 3) + 3$$

$$\hat{G}_4 = C(r+2, 4) + C(r+1, 3) + 6$$

$$\hat{G}_5 = C(r+2, 5) + C(r, 4) + C(r-1, 3) + 10$$

$$\hat{G}_6 = C(r+2, 6) + C(r, 5) + C(r-2, 4) + C(r-3, 3) + 21$$

$$\hat{G}_7 = C(r+2, 7) + C(r, 6) + C(r-2, 5) + C(r-4, 4) + C(r-5, 3) + 120$$

$$\hat{G}_i = C(r+2, i) + C(r, i-1) + C(r-2, i-2) + C(r-4, i-3)$$

$$+ C(r-5, i-4) + C(23, i-5) \quad \text{for } 8 \leq i \leq r-2.$$

(The junk constants are binomials: $3 = C(3,2)$, $6 = C(4,2)$, $10 = C(5,2)$, $21 = C(7,2)$, $120 = C(16,2)$; the corrector $C(23, i-5)$ vanishes for $i \geq 29$ — “automatic taper”.) For $8 \leq i \leq r-2$ the displayed form is the canonical digit sequence of $\hat{G}_i$: digits $r+2 > r > r-2 > r-4 > r-5 > 23$ (strictly decreasing since $r \geq 29$) at consecutive positions $i \dots i-5$, digits $\geq$ positions $(r-5 \geq i-4 \iff i \leq r-1; 23 \geq i-5 \text{ for } i \leq 28, \text{ and for } i \geq 29 \text{ the term is } 0 \text{ and the sequence ends at position } i-4)$. Levels 3-7 are likewise canonical (junk digits 3, 4, 5, 7, 16 at position 2, all below the digit above them: $16 < r-5$ needs $r \geq 22$).

Lemma RS (reverse steps). For every $3 \leq i \leq r-2$:

$$r + \partial_i(\hat{G}_i + 1) > \hat{G}_{i-1}.$$

Proof. Each case is obtained by termwise shadowing of the displayed canonical cascade for $\hat{G}_{i+1}$; throughout we use $C(n,k) - C(n-1,k) = C(n-1,k-1)$ and $C(m,2) - C(m-1,2) = m-1$.

$i = 3$: $\hat{G}_{3+1} = C(r+3,3)+C(3,2)+C(1,1)$, shadow $C(r+3,2)+3+1$, so $\text{LHS} = C(r+3,2)+r+4 = C(r+4,2)+1 = \hat{G}_2 + 1$. [difference 1]

$i = 4$: $\hat{G}_{4+1} = C(r+2,4)+C(r+1,3)+C(4,2)+C(1,1)$, shadow $C(r+2,3)+C(r+1,2)+4+1$; $\text{LHS} - \hat{G}_3 = r + C(r+1,2)+5 - C(r+2,2) - 3 = r + 2 - (r+1) = 1$. [difference 1]

$i = 5$: $\hat{G}_{5+1} = C(r+2,5)+C(r,4)+C(r-1,3)+C(5,2)+C(1,1)$, shadow $C(r+2,4)+C(r,3)+C(r-1,2)+5+1$; $\text{LHS} - \hat{G}_4 = r + C(r,3)+C(r-1,2)+6 - C(r+1,3) - 6 = r + C(r-1,2) - C(r,2) = 1$.

$i = 6$: $\hat{G}_{6+1} = C(r+2,6)+C(r,5)+C(r-2,4)+C(r-3,3)+C(7,2)+C(1,1)$ ($22 = C(7,2)+C(1,1)$), shadow $\dots+C(r-3,2)+7+1$; $\text{LHS} - \hat{G}_5 = r + C(r-2,3)+C(r-3,2)+8 - C(r-1,3) - 10 = r - 2 + C(r-3,2) - C(r-2,2) = r - 2 - (r-3) = 1$.

$i = 7$: $\hat{G}_{7+1} = C(r+2,7)+C(r,6)+C(r-2,5)+C(r-4,4)+C(r-5,3) + C(16,2)+C(1,1)$, shadow $\dots+C(r-5,2)+16+1$; $\text{LHS} - \hat{G}_6 = r + C(r-4,3)+C(r-5,2)+17 - C(r-3,3) - 21 = r - 4 + C(r-5,2) - C(r-4,2) = r - 4 - (r-5) = 1$.

$i = 8$ (the switch): $\hat{G}_{8+1} = C(r+2,8)+C(r,7)+C(r-2,6)+C(r-4,5) + C(r-5,4)+C(23,3)+C(2,2)$, shadow $C(r+2,7)+C(r,6)+C(r-2,5)+C(r-4,4)+C(r-5,3)+C(23,2)+C(2,1)$; $\text{LHS} - \hat{G}_7 = r + 253 + 2 - 120 = r + 135$.

$9 \leq i \leq 28$: $\hat{G}_{i+1}$ appends the unary digit $i-6$ at position $i-6$ (valid: $i-6 < 23$), so the shadow is [the five leading terms shifted] + $C(23, i-6) + (i-6) = \hat{G}_{i-1} + (i-6) - [0]$, giving difference $r + i - 6$.

$i = 29$: this case is nonempty only when $r \geq 31$. Since $C(23,24) = 0$, $\hat{G}_{29+1}$ appends the unary digit 24 at position 24 (valid: $24 < r-5$), and the shadow is [five terms shifted] + $C(24,23) = (\hat{G}_{28} - C(23,23)) + 24 = \hat{G}_{28} + 23$; difference $r + 23$.

$30 \leq i \leq r-2$: all corrector terms have vanished. Thus $\hat{G}_{i+1}$ appends the unary digit $i-5$ at position $i-5$ (valid: $i-5 < r-5$), and its shadow is [the five leading terms shifted] + $(i-5) = \hat{G}_{i-1} + (i-5)$. The difference is therefore exactly $r+i-5$. $\square$

Lemma IND (the bound). If the profile is feasible then $c_i \leq \hat{G}_i$ for every $2 \leq i \leq r-2$. *Proof.* $c_2 \leq C(A,2) = \hat{G}_2$. Inductively, if $c_{i-1} \leq \hat{G}_{i-1}$ and $c_i \geq \hat{G}_i + 1$, then by monotonicity $c_{i-1} = r + \partial_i(c_i) \geq r + \partial_i(\hat{G}_i + 1) > \hat{G}_{i-1}$, a contradiction; so $c_i \leq \hat{G}_i$. $\square$

2. The forced top of the chain (Lemma L4-T)

Exact computation of three steps of the chain by termwise shadowing of the displayed canonical cascades, for $r \geq 15$:

- $c_{\{r+1\}} = r$: unary digits $r+1, r, \dots, 2$ at positions $r+1 \dots 2$. Shadow $\Sigma_{\{j=2\}}^{\{r+1\}} j = C(r+2, 2) - 1$, so $\mathbf{c}_r = \mathbf{C}(r+2, 2) + (r-1)$, with digit sequence $C(r+2, r) + [\text{unary } r-1, \dots, 1 \text{ at positions } r-1 \dots 1]$.
- Shadow of c_r : $C(r+2, r-1) + \Sigma_{\{j=1\}}^{\{r-1\}} j = C(r+2, r-1) + C(r, 2)$, so $\mathbf{c}_{\{r-1\}} = \mathbf{C}(r+2, r-1) + \mathbf{C}(r, 2) + r$, with digit sequence $C(r+2, r-1) + C(r, r-2) + C(r-2, r-3) + C(r-4, r-4) + C(r-5, r-5)$ (indeed $C(r, r-2) = C(r, 2)$, then $r = (r-2) + 1 + 1$).
- Shadow: $C(r+2, r-2) + C(r, r-3) + C(r-2, r-4) + (r-4) + (r-5)$, so $\mathbf{c}_{\{r-2\}} = \mathbf{C}(r+2, 4) + \mathbf{C}(r, 3) + \mathbf{C}(r-2, 2) + (3r - 9)$ in value (positions $r-2, r-3, r-4$ for the three leading terms). $\square$

(These three identities are re-verified numerically for every r in the verification range; the verifier recomputes the chain with a separately implemented exact `kk_shadow`.)

3. The contradiction

For $r \geq 31$, $C(23, (r-2)-5) = C(23, r-7) = 0$ ($r-7 > 23$), so

$$\begin{aligned} \hat{G}_{\{r-2\}} &= C(r+2, 4) + C(r, 3) + C(r-2, 2) \\ &\quad + C(r-4, r-5) + C(r-5, r-6) \\ &= C(r+2, 4) + C(r, 3) + C(r-2, 2) + (2r-9). \end{aligned}$$

Indeed, $C(r-4, r-5) = r-4$ and $C(r-5, r-6) = r-5$. Hence

$$c_{\{r-2\}} - \hat{G}_{\{r-2\}} = (3r - 9) - (2r - 9) = r > 0,$$

contradicting Lemma IND at $i = r-2$. For $r \in \{29, 30\}$ the corrector term $C(23, r-7)$ equals $C(23, 22) = 23$ resp. $C(23, 23) = 1$ and the margin is $r - 23 = 6$ resp. $r - 1 = 29$, still positive. Therefore the profile is infeasible and $\alpha_r \leq r - 1$ for all $r \geq 29$. $\square$

4. Remarks

1. **Uniformity.** Unlike the lower half, there is no upper threshold: the forward chain is used only three levels deep, so no top-zone budget arises. L4 is proved for ALL $r \geq 29$.
2. **Reverse-step margins.** The quantity $r + \partial_i(\hat{G}_i + 1) - \hat{G}_{i-1}$ equals 1 for $i = 3, \dots, 7$, equals $r + 135$ at $i = 8$, and has the exact positive values displayed above thereafter. The final contradiction margin is positive (and equals r for $r \geq 31$). Thus all tight cases are confined to the six explicit head computations above.
3. **Verification.** `verification/verify_l4.py` checks the exact finite base $11 \leq r < 29$ and recomputes every reverse step, the three top identities, and the final margin at representative values through $r = 1000$. The proof above is symbolic and does not depend on any stress test.

5. Consequences (dependency update)

With L4 proved for $r \geq 29$, and the finite base checked for $11 \leq r < 29$:

- L5 follows immediately. Its two top singleton loads induce cumulative load $r + 3$ at level $r + 1$, larger than the L4-admissible maximum $r - 1$.
- The one-seed residual recurrence used in the full-profile obstruction dominates the L4 recurrence by monotonicity.
- Together with the uniform lower-window theorem in `UNIFORM_THEOREM.md`, this supplies the upper half of the core window for every $r \geq 11$.

Part IV: the uniform theorem for $r \geq 11$

A uniform threshold for the Erdős-Trotter antichain problem

Status: end-to-end Lean formalized for $r \geq 4$ after four separate adversarial AI review passes; pending independent specialist peer review **Date:** 2026-07-17

Throughout this note, $n_0(r)$ denotes the least integer N such that a target family exists for every $n > N$. Thus the strict inequality is part of the convention: the displayed value is the last failing ground-set size.

Main result of this note:

$$n_0(r) = 2r + 5 \quad (r \geq 11).$$

The conceptual interface is the upper-core carry. Put $A = r + 4$, let a_b be the constant upper-core recurrence, and prove

$$a_b \leq \binom{A}{b} \quad (4 \leq b \leq r + 2), \quad a_3 = \binom{A}{3} + e, \quad 1 \leq e \leq 6. \quad (0.1)$$

Lower-window feasibility gives the capacities and $e \leq 6$; L4 gives $e \geq 1$. Positive carry gives the obstruction, bounded carry gives the feasible core, puncturing gives free sets, and padding gives every later ground-set size.

For later interpretation, define α_r to be the largest integer y for which the profile

$$f_i = r \quad (2 \leq i \leq r), \quad f_{r+1} = y$$

is feasible on an $A = r + 4$ point ground set. Equivalently, α_r is the largest y for which the recurrence

$$m_{r+1} = y, \quad m_i = r + \partial_{i+1}(m_{i+1}) \quad (i = r, \dots, 2)$$

satisfies

$$m_i \leq \binom{A}{i} \quad (2 \leq i \leq r + 1).$$

The equivalence follows from the prescribed-profile criterion. Part I proves $\alpha_r \geq r - 6$, and the carry identity implies $\alpha_r = r - e$, but α_r is not needed as an object in the main logical dependency.

The point of this proof is that the old top-zone cutoff $r \leq 28\,710$ disappears. The replacement is:

1. follow the exact anchor ladder as far as it naturally goes;
2. complete one additional anchor;
3. build a short sequence of compressed phases;
4. use a one-digit rising tail to enter an exact generalized envelope;
5. let a small scalar recurrence perform the entire bottom collapse.

The human proof uses two finite exact checks: the lower-window base $11 \leq r \leq 377$, consisting of 367 integer cascade computations, and the L4 base $11 \leq r \leq 28$, consisting of 18 exact profile checks. The lower window is symbolic for $r \geq 378$, while L4 is symbolic for $r \geq 29$. Additional witness checks and adversarial computations archived with this repository are supplementary falsification tests. In Lean, the range $11 \leq r \leq 28$ is discharged by

a direct bounded upper-core/overflow certificate, $29 \leq r \leq 377$ by a bounded lower-window certificate, and $r \geq 378$ symbolically. The resulting least-threshold theorem is formalized for every $r \geq 4$.

1. Kruskal-Katona notation

For $m \geq 0$ and $k \geq 1$, write the k -cascade of m as

$$m = \binom{a_k}{k} + \binom{a_{k-1}}{k-1} + \cdots + \binom{a_s}{s}, \quad a_k > a_{k-1} > \cdots > a_s \geq s.$$

Set

$$\partial_k(m) = \binom{a_k}{k-1} + \binom{a_{k-1}}{k-2} + \cdots + \binom{a_s}{s-1}, \quad \partial_k(0) = 0.$$

For lower bounds f , define

$$m_s = f_s, \quad m_i = f_i + \partial_{i+1}(m_{i+1}),$$

The profile theorem is used asymmetrically:

- any antichain with at least f_i members at level i forces

$$m_i \leq \binom{n}{i}$$

at every level;

- if all capacities fit, the colex construction produces an antichain with exactly f_i members at every requested level.

Thus obstruction never requires trimming, while construction obtains the exact profile.

A self-contained necessity-and-sufficiency proof, together with the canonical colex construction, is given in [TECHNICAL_NOTE.md, Section 2](#).

We repeatedly use two elementary facts.

Monotonicity

If $x \leq y$, then

$$\partial_k(x) \leq \partial_k(y).$$

Completion of a constant-offset run

A finite cascade run

$$\binom{d}{p} + \binom{d-1}{p-1} + \cdots + \binom{d-\ell}{p-\ell}$$

is bounded above by the corresponding completed binomial

$$\binom{d+1}{p}.$$

This is just Pascal/hockey-stick expansion. Consequently, after a run is shadowed, it may be rounded up to its completed stone in a supersolution.

A **supersolution** for the recurrence is a sequence V_i satisfying

$$V_{i-1} \geq r + \partial_i(V_i).$$

If the exact recurrence is below V_i at one level, monotonicity keeps it below the supersolution at every lower level.

Notation roadmap

The proof has two long recurrences and one phased supersolution. The following table fixes the recurring symbols and, in particular, separates phase positions from the residual used in the final obstruction.

Symbol	Meaning
$A = r + 4$	ground-set size for the lower window and reflected upper core
$B = 2r + 4$	diagonal-shell parameter for the obstruction on $B + 1 = 2r + 5$ points
m_i	exact lower-window recurrence state in (2.1)
T_q	anchor threshold defined recursively in (3.1)
$B_q = T_q - 2q$	bottom digit of the unary block at anchor q
t	maximal anchor index with $T_t \leq r < T_{t+1}$
N, c, h	phase parameters $N = r - 2t$, $c = 2t + 3$, and $h = t + 2$
K	final phase index selected by (5.1)
s_j	tower stone $r + 1 - c2^{j-1}$
b_j, u_j, ℓ_j	birth level, active top position, and extension count for phase j
a_i	reflected upper-core recurrence in Sections 14-16
M_i	full-profile recurrence on $2r + 5$ points in Section 16
ν_i	residual recurrence after the diagonal shell is removed
c_i	L4 recurrence used to lower-bound ν_i

Part I. The uniform lower window

2. The lower-window recurrence

Put

$$A = r + 4.$$

To prove $\alpha_r \geq r - 6$, it is enough to show that the recurrence

$$m_{r+1} = r - 6, \quad m_i = r + \partial_{i+1}(m_{i+1}) \quad (i = r, r - 1, \dots, 2) \quad (2.1)$$

satisfies

$$m_i \leq \binom{A}{i} \quad (2 \leq i \leq r + 1). \quad (2.2)$$

The range $11 \leq r \leq 377$ is checked exactly in Section 12. Hence assume throughout Sections 3–11 that

$$r \geq 378.$$

Conceptual segment table

The detailed proof below is one supersolution assembled from the following segments. The full theorem/module cross-reference is in [PROOF_GUIDE.md](#).

Segment, levels, and state	Transition, capacity, and Lean modules
Exact anchors: $r + 1$ to $r - t$; (3.2)	Lemma 3.1 and the exact top-segment bound. Lean: Anchors.
Completed anchor: $r - t - 1$; (4.1)–(4.2)	Rounded-anchor and phase-one entry; completed-anchor bound. Lean: Anchors, Phase Start.
Ordinary phases: phase-birth intervals; (6.0)	Extension and harvest lemmas; Lemma 5.1 and (6.2), (6.4). Lean: Phase Canonical, Transitions, Harvest, Capacity.
Rising tail: final harvest to level L ; (7.2)–(7.3)	Steps (7.4)–(7.5) and rising-state capacity. Lean: Rising Tail.
Generalized envelope: $t + K + 4$ to $t + 5$; (8.1)	Exact step (8.2) and envelope capacity. Lean: Generalized Envelope.
Scalar collision: $t + 5$ to $t + 4$; E_i to $F_t(0)$	Collision identity in Section 9 and endpoint capacity. Lean: Scalar States, Full Envelope.
Scalar collapse: $t + 4$ to 4 ; (9.1)	Lemmas 10.1–10.2 and scalar-state capacity. Lean: Scalar Collapse, Scalar States.
Final levels: $4, 3, 2$; final scalar states	Equations (11.1)–(11.3), including tight bottom capacity. Lean: Final Levels, Phase Assembly.

3. Exact anchor ladder

Define

$$T_0 = 28, \quad T_{q+1} = \binom{T_q - 2q}{2}, \quad (3.1)$$

and put

$$B_q = T_q - 2q.$$

Thus

$$T_1 = 378, \quad T_2 = 70\,500, \quad T_3 = 2\,484\,807\,760,$$

and so on.

For integers $a \leq b$, write

$$U(a, b) = \sum_{x=a}^b \binom{x}{x},$$

with $U(a, b) = 0$ if $a > b$.

Lemma 3.1: exact anchors

If $r \geq T_q$, then the exact recurrence (2.1) satisfies

$$m_{r-q} = \sum_{j=0}^q \binom{r+2-2j}{r-q-j} + U(B_q, r-2q-1). \quad (3.2)$$

Proof For $q = 0$, the $r - 6$ top sets are unary at level $r + 1$, so

$$m_r = \binom{r+2}{r} + U(28, r-1).$$

Assume (3.2). Let

$$N = r - 2q.$$

After taking one shadow, the unary part contributes

$$\sum_{x=B_q}^{N-1} x = \binom{N}{2} - \binom{B_q}{2}.$$

Adding the new flow r gives

$$\binom{N}{2} + (r - T_{q+1}).$$

The first term is the new anchor

$$\binom{N}{N-2},$$

and, when $r \geq T_{q+1}$, the residual $r - T_{q+1}$ is the unary block

$$U(T_{q+1} - 2q - 2, r - 2q - 3) = U(B_{q+1}, r - 2(q+1) - 1).$$

This proves the induction. $\square$

Capacity on the exact top segment

The exact states just constructed already satisfy the required capacity bounds. At the top level,

$$m_{r+1} = r - 6 < \binom{r+4}{r+1}.$$

At level $i = r - q$ in Lemma 3.1, the canonical cascade has leading digit $r + 2$ at position i , and every remaining digit is strictly smaller. The remaining positions form a canonical cascade of index at most $i - 1$ whose digits are at most $r + 1$, so its value is strictly less than $\binom{r+2}{i-1}$. Consequently

$$m_i < \binom{r+2}{i} + \binom{r+2}{i-1} = \binom{r+3}{i} < \binom{r+4}{i}.$$

Thus every level from $r + 1$ down through the last exact anchor level fits on the $A = r + 4$ -point ground set.

Choose $t \geq 1$ maximal such that

$$T_t \leq r < T_{t+1}. \quad (3.3)$$

Set

$$N = r - 2t, \quad c = 2t + 3, \quad h = t + 2. \quad (3.4)$$

A useful elementary bound is

$$T_t \geq 128t + 250 \quad (t \geq 1). \quad (3.5)$$

It is equality at $t = 1$, and the induction step follows immediately from (3.1). In particular,

$$r \geq 64c. \quad (3.6)$$

4. Complete one additional anchor

At level

$$I = r - t,$$

Lemma 3.1 gives the exact state. Descending one more level, the residual below the old anchors is

$$\binom{N}{2} - (T_{t+1} - r).$$

Since $r < T_{t+1}$, this is at most $\binom{N}{2}$. It is also nonnegative: since $r \geq T_t$,

$$N = r - 2t \geq T_t - 2t = B_t,$$

and therefore

$$\binom{N}{2} - (T_{t+1} - r) = \binom{N}{2} - \binom{B_t}{2} + r \geq 0.$$

We therefore round it up and define

$$V_{I-1} = \sum_{j=0}^{t+1} \binom{r+2-2j}{I-1-j}. \quad (4.1)$$

This is a valid supersolution step.

The new final anchor is N , and after one shadow it sits at position $N - 3$. At the next level

$$b_1 = I - 2 = r - t - 2$$

place beneath it the two-term run

$$\binom{N-3}{N-4} + \binom{N-4}{N-5}. \quad (4.2)$$

Its value is

$$2N - 7 \geq r \quad (4.3)$$

by (3.5). Thus (4.2) pays the next flow.

Define the tower stones

$$\boxed{s_j = r + 1 - c 2^{j-1} \quad (j \geq 1).} \quad (4.4)$$

Then

$$s_1 = N - 2,$$

and the run in (4.2) has top digit $s_1 - 1$. Moreover

$$s_{j+1} = 2s_j - r - 1. \quad (4.5)$$

5. Choice of the final phase

Let K be the unique integer satisfying

$$c 2^{K-1} \leq \frac{r}{4} < c 2^K. \quad (5.1)$$

Because of (3.6), $K \geq 5$.

For $1 \leq j \leq K$,

$$s_j \geq \frac{3r}{4} + 1, \quad (5.2)$$

and

$$s_{K+1} \geq \frac{r}{2} + 1. \quad (5.3)$$

The first phase begins with top position

$$u_1 = N - 4. \quad (5.4)$$

It already has two terms, as in (4.2).

For the first phase, set

$$\ell_1 = \left\lfloor \frac{u_1 - 5}{2} \right\rfloor. \quad (5.5)$$

After ℓ_1 extensions, the run bottom is 4 or 5. Harvest the run to s_1 and birth phase 2.

For every later ordinary phase $j \geq 2$, if its top position is u_j , set

$$\ell_j = \left\lfloor \frac{u_j - 4}{2} \right\rfloor, \quad u_{j+1} = u_j - \ell_j - 2 = \left\lceil \frac{u_j}{2} \right\rceil. \quad (5.6)$$

The first transition differs by at most one:

$$u_2 \leq \left\lceil \frac{u_1}{2} \right\rceil + 1.$$

Let b_j be the birth level of phase j . Then

$$b_1 = I - 2, \quad b_{j+1} = b_j - \ell_j - 1, \quad (5.6a)$$

and

$$\boxed{b_j - u_j = h + j - 1 = t + j + 1.} \quad (5.6b)$$

The identity holds at $j = 1$, because $(I - 2) - (N - 4) = t + 2 = h$. A harvest decreases the level by $\ell_j + 1$ and the next active top position by $\ell_j + 2$, so the slot increases by exactly one. In particular, the phases cover every intervening level without gaps.

Consequently

$$\frac{u_1}{2^{j-1}} \leq u_j \leq \frac{u_1}{2^{j-1}} + 2. \quad (5.7)$$

For completeness, the lower bound follows because every transition is at least halving. For the upper bound, the exceptional first transition gives $u_2 \leq u_1/2 + 3/2 \leq u_1/2 + 2$. If the upper bound holds at some $j \geq 2$, then

$$u_{j+1} = \left\lceil \frac{u_j}{2} \right\rceil \leq \frac{u_1}{2^j} + \frac{3}{2} \leq \frac{u_1}{2^j} + 2,$$

which proves the two-case induction. Using (5.1), (5.7), and $u_1 > 3r/4$, we get

$$\boxed{3c \leq u_K \leq 8c + 2.} \tag{5.8}$$

Lemma 5.1: quantitative phase bounds

The estimates used in Sections 6–8 follow uniformly from the preceding parameters:

1. $3c \leq u_K \leq 8c + 2$.
2. Every phase-1 extension term in (6.1) has lower index at least 4, complementary index at least 2, and upper digit at least $(N - 1)/2$.
3. Every later ordinary extension term in (6.3) has lower index at least 4, complementary index at least 2, and upper digit at least $r/2 - 3$.
4. Every newborn term at an ordinary harvest has upper digit at least $3r/4$, lower index at least 2, and complementary index at least 2; hence it has value greater than r .
5. In the final rising-tail zone, $d_p \geq 3r/8$ whenever $p \geq 4$.
6. The terminal envelope level satisfies $L < r/4$.

Proof. From $u_1 = r - c - 1$, (5.1), and (5.7),

$$u_K \geq \frac{r - c - 1}{2^{K-1}} \geq 4c - \frac{4c(c+1)}{r} > 3c,$$

where the last inequality uses $r \geq 64c$. The other half of (5.1) gives $2^{-(K-1)} < 8c/r$, hence

$$u_K \leq \frac{u_1}{2^{K-1}} + 2 < 8c + 2.$$

For phase 1, the extension in (6.1) has index $u_1 - 3 - 2\ell$, which is at least 4 because $\ell < \ell_1$. Its complementary index is

$$(s_1 - 3 - \ell) - (u_1 - 3 - 2\ell) = 2 + \ell,$$

and the endpoint choice of ℓ_1 gives upper digit at least $(N - 1)/2$.

For a later phase, the index in (6.3) is at least 4. Its complementary index equals

$$s_j - u_j + \ell.$$

For $2 \leq j \leq K - 1$, we have $c2^{j-1} \leq r/8$ and $u_j \leq r/2 + 2$, so

$$s_j - u_j \geq r + 1 - \frac{r}{8} - \left(\frac{r}{2} + 2\right) > 2.$$

Also $\ell \leq u_j/2$, and therefore

$$s_j - 2 - \ell \geq \frac{3r}{4} - 1 - \frac{r/2 + 2}{2} \geq \frac{r}{2} - 3.$$

At harvest, the next stone has digit at least $3r/4 + 1$, so the newborn active digit is at least $3r/4$.

Its position is at most $r/2 + 2$, and the preceding bounds leave both index and complementary index at least 2; its value is therefore at least $\binom{3r/4}{2} > r$.

In the final zone, $D \geq r/2 + 3$ and $p \leq u - 1 \leq r/8 + 1$, so

$$d_p = D - p + 1 \geq \frac{3r}{8} + 3 > \frac{3r}{8}.$$

Finally, (3.5) gives $t < r/128$, while (5.1) gives $K \leq \log_2 r$. Thus

$$L = t + K + 4 < \frac{r}{128} + \log_2 r + 4 < \frac{r}{4} \quad (r \geq 378).$$

□

6. Ordinary phase transitions

The full active-run invariant is as follows. At level $b_1 - L$, after L completed phase-1 extensions, the active run is

$$R_{1,L} = \sum_{p=0}^{L+1} \binom{s_1 - 1 - p}{u_1 - L - p}, \quad 0 \leq L \leq \ell_1. \quad (6.0a)$$

For every later phase $j \geq 2$, at level $b_j - L$, after L completed extensions, the active run is

$$R_{j,L} = \sum_{p=0}^L \binom{s_j - 1 - p}{u_j - L - p}, \quad 0 \leq L \leq \ell_j. \quad (6.0b)$$

The completed anchors and stones simply shadow one position at a time. The digits and positions in each run both decrease by one as p increases, so the displayed runs are canonical. After one further shadow, the hockey-stick completion dominates the old run, and the newborn term sits at $u_j - \ell_j - 2 = u_{j+1}$. These formulas fix the meanings of “extension”, “harvest”, and “birth” below.

Worked schematic: one ordinary extension

Fix a phase $j \geq 2$ and suppose $0 \leq L < \ell_j$. At level $b_j - L$ its active run is

$$R_{j,L} = \sum_{p=0}^L \binom{s_j - 1 - p}{u_j - L - p}.$$

Termwise shadowing lowers every position by one:

$$\partial_{b_j - L}(R_{j,L}) = \sum_{p=0}^L \binom{s_j - 1 - p}{u_j - L - p - 1}.$$

The next prescribed load is paid by appending the new final digit

$$Q_{j,L} = \binom{s_j - L - 2}{u_j - 2L - 2}.$$

The bounds in Lemma 5.1 ensure that this digit is valid, lies below the preceding digit, and satisfies $Q_{j,L} \geq r$. Consequently

$$r + \partial_{b_j-L}(R_{j,L}) \leq \partial_{b_j-L}(R_{j,L}) + Q_{j,L} = R_{j,L+1}.$$

All completed anchors and earlier stones shadow unchanged alongside this calculation. This is the exact transition formalized by `phase_extension_pays` and `isCanonical_phase_extension`. At $L = \ell_j$, the same run is completed by the hockey-stick identity; the harvest transition places the next active digit at $u_j - \ell_j - 2 = u_{j+1}$ and begins phase $j + 1$. Thus one extension and one harvest account for every seam used in the ordinary-phase segment.

Phase 1

At birth level $b_1 = I - 2$, phase 1 consists of

$$s_1 - 1, s_1 - 2$$

at positions

$$u_1, u_1 - 1.$$

For $0 \leq \ell < \ell_1$, shadow to level $b_1 - 1 - \ell = I - 3 - \ell$ and append

$$\binom{s_1 - 3 - \ell}{u_1 - 3 - 2\ell}. \quad (6.1)$$

Whenever this extension is used, its lower index is at least 4, its complement is at least 2, and its upper digit is at least

$$\frac{N - 1}{2}.$$

From (3.5), $t \leq (r - 250)/128$, so

$$N = r - 2t \geq \frac{63r}{64} + \frac{125}{32} > \frac{r}{2}.$$

Thus the upper digit is larger than $r/4 - 1$; for $r \geq 378$, even the smaller endpoint binomial $\binom{\lfloor r/4 \rfloor - 1}{2}$ exceeds r . For a fixed upper digit, a binomial coefficient whose index and complementary index are both at least 2 is at least its endpoint value $\binom{d}{2}$. Since the actual lower index is at least 4 and the complementary index is at least 2, we obtain

$$\binom{s_1 - 3 - \ell}{u_1 - 3 - 2\ell} \geq r. \quad (6.2)$$

Phases 2, ..., $K - 1$

At birth level b_j , phase j begins with

$$\binom{s_j - 1}{u_j}.$$

For $0 \leq \ell < \ell_j$, shadow to level $b_j - 1 - \ell$ and append

$$\binom{s_j - 2 - \ell}{u_j - 2\ell - 2}. \quad (6.3)$$

For $j \leq K - 1$, (5.2), (5.7), and (5.8) imply that throughout every extension:

- the lower index is at least 4;
- the complementary index is at least 2;
- the upper digit is at least $r/2 - 3$.

Hence

$$\binom{s_j - 2 - \ell}{u_j - 2\ell - 2} \geq \binom{r/2 - 3}{2} > r. \quad (6.4)$$

At a harvest step, completion of the old run costs nothing: the completed stone dominates its shadow. The newly born active term for the next phase has upper digit at least $3r/4$, while its position lies between 2 and $r/2 + 2$. Therefore that newborn term alone has value greater than r .

Thus every ordinary in-phase and harvest transition is a valid supersolution step.

7. A one-digit final zone

At the harvest of phase $K - 1$, namely at level b_K , do not birth an ordinary phase K . Instead, place the completed stone s_K at position u_K , and place the new tail immediately below it at position $u_K - 1$. Completion of the old phase- $(K - 1)$ run dominates its shadow, while

$$\binom{s_K}{u_K} \geq \binom{3r/4}{2} > r.$$

Thus the new s_K -term alone pays the incoming flow; the tail introduced below it is additional slack.

Put

$$u = u_K, \quad D = s_{K+1} + 2. \quad (7.1)$$

At the position immediately below s_K , namely $u - 1$, place the digit

$$d_{u-1} = D - u + 2. \quad (7.2)$$

As the construction descends, let this one tail digit move from position p to $p - 1$, while changing according to

$$d_p = \begin{cases} D, & p = 2, \\ D - p + 1, & p \geq 3. \end{cases} \quad (7.3)$$

Thus the digit rises by 1 at every step except the final transition $3 \rightarrow 2$, where it rises by 2.

Canonical validity

From (5.1),

$$s_K - D = c2^{K-1} - 2 > 0.$$

Also, by (5.3), (5.8), and (3.6),

$$D \geq \frac{r}{2} + 3, \quad u \leq \frac{r}{8} + 2.$$

Hence

$$D - u + 2 \geq u - 1,$$

so the initial tail term is valid, and it remains strictly below s_K .

Gain at each step

For $p \geq 4$, the gain from raising by one is

$$\binom{d_p + 1}{p - 1} - \binom{d_p}{p - 1} = \binom{d_p}{p - 2}. \quad (7.4)$$

Here

$$d_p \geq \frac{3r}{8}, \quad 2 \leq p - 2 \leq d_p - 2,$$

so (7.4) is greater than r .

For the last step,

$$\binom{D}{2} - \binom{D - 2}{2} = 2D - 3 \geq r. \quad (7.5)$$

Therefore the rising tail pays every remaining flow.

After $u - 3$ steps, s_K has reached position 3, and the tail has reached position 2 with digit $D = s_{K+1} + 2$. The level reached is

$$b_K - (u_K - 3) = (b_K - u_K) + 3 = h + K + 2. \quad (7.5a)$$

More generally, once s_q is completed, its permanent slot is

$$\text{level} - \text{position} = h + q - 1. \quad (7.5b)$$

This is the simultaneous alignment invariant at the Section 7/8 seam.

Let

$$L = h + K + 2. \quad (7.6)$$

The state at level L is exactly the generalized envelope introduced next.

8. Exact generalized envelope

For $h + 3 \leq i \leq L$, define

$$E_i = \sum_{j=0}^{h-1} \binom{r + 2 - 2j}{i - j} + \sum_{q=1}^{i-h-2} \binom{s_q}{i - h - q + 1} + \binom{s_{i-h-1} + 2}{2}. \quad (8.1)$$

At level L , this is exactly the endpoint of Section 7. Indeed, by (7.5b), stone s_q occupies position $L - (h + q - 1) = L - h - q + 1$, exactly its position in the second sum of (8.1); s_K is at position 3, and the tail $s_{K+1} + 2$ is at position 2.

The representation is canonical:

- the initial anchors fall by 2;
- $r - 2t = N > s_1 = N - 2$;
- successive tower stones satisfy $s_q > s_{q+1}$;
- the terminal digit $s_{i-h-1} + 2$ is still below the preceding stone.

Moreover, all tower digits used here are at least $s_{K+1} \geq r/2 + 1$. From (3.5), $t < r/128$; from

(5.1), $K \leq \log_2 r$. Hence

$$L = t + K + 4 < \frac{r}{128} + \log_2 r + 4 < \frac{r}{4} \quad (r \geq 378),$$

so every digit is above its position.

Exact envelope step

The fixed anchors and old stones simply shift under one shadow. At the bottom, write

$$m = i - h - 1.$$

The difference needed to pass from E_i to E_{i-1} is

$$\binom{s_{m-1} + 2}{2} - \binom{s_{m-1}}{2} - (s_m + 2).$$

Using (4.5), this equals

$$2s_{m-1} - s_m - 1 = r.$$

Hence

$$\boxed{E_{i-1} = r + \partial_i(E_i) \quad (h + 4 \leq i \leq L).} \quad (8.2)$$

9. Collision of the extra anchors

At level

$$i = h + 3 = t + 5,$$

the last anchor is N at position 4, the first tower stone is

$$s_1 = N - 2$$

at position 3, and the terminal digit is $s_2 + 2$ at position 2.

After one shadow and addition of r , the part below the shifted N anchor is

$$\binom{N-2}{2} + (s_2 + 2) + r.$$

Since

$$s_2 + 2 = r - 4t - 3$$

and

$$\binom{N}{2} - \binom{N-2}{2} = 2N - 3 = 2r - 4t - 3,$$

this part is exactly $\binom{N}{2}$. It merges with $\binom{N}{3}$ to give $\binom{N+1}{3}$.

Thus, at level $t + 4$, the state is

$$F_t(0),$$

where for $0 \leq q \leq t$ and $z \geq 0$ we define

$$F_q(z) = \sum_{j=0}^q \binom{r+2-2j}{q+4-j} + \binom{r-2q+1}{3} + z, \quad (9.1)$$

with z inserted in its 2-cascade below the displayed position-3 term.

10. Scalar bottom collapse

Set

$$z_t = 0,$$

and for $q = t, t-1, \dots, 1$, define

$$z_{q-1} = 2q - 1 + \partial_2(z_q). \quad (10.1)$$

The residuals really do append canonically below the position-3 digit. For $q \geq 6$, the proof of Lemma 10.2 gives $z_q \leq q^2$, so the top digit of the 2-cascade of z_q is at most $2q$. For the smaller indices the same proof gives

$$z_5 \leq 20, \quad z_4 \leq 16, \quad z_3 \leq 14, \quad z_2 \leq 11, \quad z_1 \leq 9. \quad (10.1a)$$

Since $\binom{7}{2} = 21$, their top 2-cascade digit is at most 6. On the other hand, by (3.5),

$$r - 2q + 1 \geq r - 2t + 1 > 2q.$$

Thus no residual carry enters the displayed position-3 digit.

Lemma 10.1

For every $q \geq 1$,

$$r + \partial_{q+4}(F_q(z_q)) = F_{q-1}(z_{q-1}). \quad (10.2)$$

Proof At level $q+4$, the last ordinary anchor has digit

$$a = r - 2q + 2$$

at position 4, and the special digit is $a-1$ at position 3. After shadowing, filling the position-2 term from $\binom{a-1}{2}$ to $\binom{a}{2}$ costs

$$a - 1 = r - 2q + 1.$$

The incoming amount is

$$r + \partial_2(z_q).$$

The excess is therefore

$$r + \partial_2(z_q) - (r - 2q + 1) = 2q - 1 + \partial_2(z_q) = z_{q-1}.$$

Finally,

$$\binom{a}{3} + \binom{a}{2} = \binom{a+1}{3},$$

which is precisely the new special digit in F_{q-1} . $\square$

Lemma 10.2: terminal residual

The recurrence (10.1) gives

$$z_0 = \begin{cases} 1, & t = 1, \\ 4, & t = 2, \\ 6, & t \geq 3. \end{cases} \quad (10.3)$$

Proof The cases $t = 1, 2$ are direct.

Assume $t \geq 3$. We first show

$$5 \leq z_2 \leq 11. \quad (10.4)$$

The lower bound follows from the $q = 3$ step.

For the upper bound, the first three cases are

t	$(z_t, z_{t-1}, \dots, z_0)$
3	$(0, 5, 7, 6)$
4	$(0, 7, 10, 8, 6)$
5	$(0, 9, 12, 11, 9, 6)$.

If $t \geq 6$, use the invariant

$$z_q \leq q^2 \quad (q \geq 6).$$

Indeed, if $q \geq 7$ and $z_q \leq q^2$, let a_2 be the top digit of the canonical 2-cascade of z_q . If $a_2 \geq 2q$, then $z_q \geq \binom{2q}{2} > q^2$, a contradiction. Hence $a_2 \leq 2q - 1$. A possible position-1 term contributes only one more to the shadow, so

$$\partial_2(z_q) \leq a_2 + 1 \leq 2q.$$

Therefore

$$z_{q-1} \leq 4q - 1 \leq (q - 1)^2.$$

Starting from $z_t = 0$, this yields $z_6 \leq 36$. Hence

$$z_5 \leq 11 + 9 = 20, \quad z_4 \leq 9 + 7 = 16, \quad z_3 \leq 7 + 7 = 14,$$

and then

$$z_2 \leq 5 + 6 = 11.$$

For $5 \leq z_2 \leq 11$,

$$4 \leq \partial_2(z_2) \leq 6,$$

so

$$7 \leq z_1 \leq 9.$$

Every integer from 7 through 9 has 2-shadow 5. Thus

$$z_0 = 1 + 5 = 6.$$

$\square$

11. The final two levels

At level 4, the state is

$$F_0(z_0) = \binom{r+2}{4} + \binom{r+1}{3} + z_0.$$

Descending to level 3 gives

$$E_3 = \binom{r+3}{3} + (\partial_2(z_0) - 1). \quad (11.1)$$

By Lemma 10.2,

$$E_3 = \begin{cases} \binom{r+3}{3} + 1, & t = 1, \\ \binom{r+3}{3} + 3, & t \geq 2. \end{cases} \quad (11.2)$$

Therefore

$$E_2 = r + \partial_3(E_3) = \begin{cases} \binom{r+4}{2} - 1, & t = 1, \\ \binom{r+4}{2}, & t \geq 2. \end{cases} \quad (11.3)$$

All higher constructed states have canonical leading digit $r + 2$, so they are strictly below

$$\binom{r+3}{i} < \binom{r+4}{i}.$$

The level-3 values in (11.2) also lie well below $\binom{r+4}{3}$, and (11.3) handles the tight bottom level.

By monotonicity, the exact recurrence (2.1) lies below this supersolution. This proves

$$\alpha_r \geq r - 6 \quad (r \geq 378).$$

12. Finite base $11 \leq r \leq 377$

For each r in this range, evaluate (2.1) with exact integer Kruskal-Katona cascades and check (2.2). A standalone implementation is provided as `verify_lower_window_base.py`; it does not import the original project code.

The full sweep has:

- 367 values of r ;
- no failures;
- minimum capacity slack 3, attained at level 2 for every $20 \leq r \leq 377$; its first occurrence is at $r = 20$.

Thus

$$\boxed{\alpha_r \geq r - 6 \quad (r \geq 11).} \quad (12.1)$$

The standalone verifier `verification/verify_lower_window_base.py` is archived with this release. It uses exact integer arithmetic; no SAT or search is involved.

The numerical coverage boundary is worth making explicit. The finite base ends at $r = 377$, immediately below the first symbolic anchor threshold $T_1 = 378$. The named seam checks at $r = 378, 379$ lie in the first anchor regime $t = 1$; the next regime begins only at

$$T_2 = 70\,500.$$

Thus the finite sweep and seam samples do not exercise the $t \geq 2$ phase structure. Coverage of the second anchor regime and every higher regime comes from the symbolic proof formalized in Lean, not from numerical sampling.

Lean's ground-bounded evaluator certifies the subrange $29 \leq r \leq 377$ and is proved equal to the original canonical shadow recurrence. For $11 \leq r \leq 28$, the final Lean theorem instead uses a smaller direct certificate of exactly the upper-core capacities/carry and the full-profile overflow. The all-value Python sweep remains an independently implemented exact cross-check of the human lower-window proof.

Part II. Assembly around the upper-core carry

The remainder uses the carry interface (0.1). Its dependency is:

1. lower-window feasibility gives upper capacities and $e \leq 6$;
2. L4 gives $e \geq 1$;
3. $e \leq 6$ gives a feasible core on $r + 5$ points;
4. puncturing top core members gives r free sets;
5. padding gives every $n \geq 2r + 6$;
6. $e \geq 1$ enters a no-carry diagonal shell and obstructs $n = 2r + 5$.

The identity involving α_r is retained below as an interpretation of the same carry, not as a required intermediate object.

13. The L4 upper window

The companion reverse-envelope proof in [L4_PROOF.md](#) establishes

$$\boxed{\alpha_r \leq r - 1 \quad (r \geq 11).} \tag{13.1}$$

It is symbolic for $r \geq 29$, with exact finite verification for $11 \leq r \leq 28$.

In particular, the profile

$$r \text{ sets at every level } 2, \dots, r + 1$$

on $r + 4$ points has bottom recurrence value strictly larger than $(\binom{r+4}{2})$.

L5 is immediate from L4: its two top singleton loads already induce a load $r + 3$ at level $r + 1$, which exceeds the L4-admissible maximum.

14. Core carry and feasibility on $r + 5$ points

Put

$$A = r + 4.$$

Run the recurrence for the core profile

$$r \text{ sets at every level } 2, \dots, r + 2$$

and denote its states by a_i .

Complement the feasible lower-window profile with top load $r - 6$. Its reflected recurrence agrees with the upper core at every level $i \geq 4$, so it supplies

$$a_i \leq \binom{A}{i} \quad (4 \leq i \leq A - 2). \quad (14.0)$$

At level 3, reflected feasibility says

$$(r - 6) + \partial_4(a_4) \leq \binom{A}{3}.$$

Since

$$a_3 = r + \partial_4(a_4),$$

we obtain

$$a_3 \leq \binom{A}{3} + 6.$$

On the other hand, suppose

$$a_3 \leq \binom{A}{3}.$$

Together with (14.0), the recurrence states a_i then satisfy all capacity inequalities for the truncated constant profile

$$f_i = r \quad (3 \leq i \leq r + 2)$$

on $A = r + 4$ points. By profile sufficiency, there is an antichain with r members on every one of those levels. Complementation maps the levels

$$3, \dots, r + 2$$

to

$$2, \dots, r + 1,$$

producing precisely the forbidden L4 profile. Therefore

$$a_3 > \binom{A}{3}.$$

Define the positive carry by

$$a_3 = \binom{A}{3} + e_r. \quad (14.1)$$

The two preceding bounds give

$$\boxed{1 \leq e_r \leq 6.} \quad (14.2)$$

For interpretation, repeat the reflected argument with an arbitrary top load y . Level-three feasibility is equivalent to $y \leq r - e_r$, and colex sufficiency attains every such value. Consequently

$$\alpha_r = r - e_r. \quad (14.3)$$

Thus the traditional lower and upper bounds on α_r encode precisely the same carry bounds, but later sections use (14.0)–(14.2) directly.

This reflection and carry argument is formalized in `Erdos776/Uniform/CoreCarry.lean`. The certificate-facing theorem `fullMiddleProfileThreshold_of_lowerWindowFits` takes only the

capacity inequalities (2.2); complementation, reflected recurrence identification, the bounds on e_r , and all subsequent assembly are proved symbolically in Lean.

The remaining capacities on $A + 1$ points are explicit. At level 3,

$$a_3 = \binom{A}{3} + e_r \leq \binom{A}{3} + 6 < \binom{A+1}{3}.$$

At level 2, using (14.2),

$$\begin{aligned} a_2 &= r + \partial_3 \left(\binom{A}{3} + e_r \right) \\ &= r + \binom{A}{2} + \partial_2(e_r) \\ &\leq r + \binom{A}{2} + 4 \\ &= \binom{A+1}{2}. \end{aligned}$$

For $i \geq 4$, (14.0) gives $a_i \leq \binom{A}{i} < \binom{A+1}{i}$; the level-3 and level-2 bounds were just proved. Thus the complete core profile is feasible on

$$A + 1 = r + 5$$

points.

15. Explicit free sets

Let

$$q = r + 5, \quad X = [q - 2], \quad J = \{4, \dots, q - 2\}.$$

In the canonical colex core, the first $r = q - 5$ top-level sets are

$$T_j = X \setminus \{j\}, \quad j \in J.$$

Indeed, the top state r has a cascade of r unit diagonal digits. Its colex realization consists exactly of the $(r + 2)$ -subsets of $X = [r + 3]$ obtained by deleting one of the r points in $J = \{4, \dots, r + 3\}$.

We need only the following elementary observation.

Puncturing lemma. If C is an antichain, T belongs to C , and D is a proper subset of T , then D contains no member of C .

Indeed, if S belongs to C and S is contained in D , then S is contained in T . Antichainness forces $S = T$, while properness gives T not contained in D , a contradiction.

For

$$4 \leq j \leq q - 2,$$

define

$$D_j = T_j \setminus \{1\} = [q] \setminus \{1, j, q - 1, q\}. \quad (15.1)$$

There are exactly r distinct such sets, each of size $q - 4 = r + 1$. Because 1 belongs to every displayed T_j , each D_j is a proper subset of the core member T_j . The puncturing lemma

immediately makes every D_j free. No classification of exceptional or later core members is required.

The two-point padding lemma therefore applies indefinitely. At stage t , retain the persistent free sets

$$D_j^{(t)} = D_j \cup \{x_1, \dots, x_t\},$$

where x_h is one selected point from the h -th added pair $\{x_h, y_h\}$. An original core member is excluded by freeness of D_j , and a member born at stage h contains y_h , which is absent from $D_j^{(t)}$. After t stages the core parameters are

$$(q, s) = (r + 5 + 2t, r + 2 + t).$$

The core-to-full lemma is proved in [TECHNICAL_NOTE.md, Section 4](#). Applied to the padded core, it gives a full-profile antichain for both

$$n = 2r + 6 + 2t \quad \text{and} \quad n = 2r + 7 + 2t.$$

Hence

$$g(n, r) = n - 3 \quad (n \geq 2r + 6). \quad (15.3)$$

The assembly from one suitable free core through arbitrary padding and both parities is formalized by `Erdos776.Uniform.fullMiddleProfileExists_of_suitableFreeCore` in `Erdos776/Uniform/Assembly.lean`.

The preceding explicit free-set argument is formalized in `Erdos776/Uniform/FreeCore.lean`. In particular, `hasSuitableFreeCore_of_upperCoreData` constructs the canonical core and its free family from the Section 14 upper-core data; no freeness certificate is assumed by the final symbolic assembly.

16. Obstruction at $n = 2r + 5$

The symbolic argument in this section is formalized in Lean in `Erdos776/Uniform/DiagonalObstruction.lean`. Its certificate-facing endpoint takes (14.0) and $1 \leq e_r \leq 6$ as explicit hypotheses; diagonal transfer, L4 residual domination, shell peeling, overflow, and the final antichain contradiction are then proved symbolically in Lean for $r \geq 29$.

Put

$$B = 2r + 4.$$

Let M_i be the full-profile recurrence for

$$r \text{ sets at every level } 2, \dots, 2r + 3$$

on $2r + 5 = B + 1$ points.

The inequalities in (14.0) are in fact strict. Suppose first that $5 \leq i \leq A - 2$ and

$$a_i = \binom{A}{i}.$$

Then

$$a_{i-1} = r + \partial_i(a_i) = r + \binom{A}{i-1} > \binom{A}{i-1},$$

contradicting the capacity inequality at level $i - 1$.

If instead

$$a_4 = \binom{A}{4},$$

then

$$a_3 = r + \partial_4(a_4) = r + \binom{A}{3}.$$

Comparing this with

$$a_3 = \binom{A}{3} + e_r$$

gives $e_r = r$, contradicting $e_r \leq 6 < r$. Hence

$$0 \leq a_i < \binom{A}{i} \quad (4 \leq i \leq A - 2). \quad (16.0)$$

The no-carry diagonal-shell principle

The transfer is one no-carry principle rather than a collection of unrelated calculations. If

$$t \geq 0, \quad 1 \leq b < A, \quad 0 \leq x < \binom{A}{b},$$

then

$$\partial_{b+t} \left(\binom{A+t}{b+t} - \binom{A}{b} + x \right) = \binom{A+t}{b+t-1} - \binom{A}{b-1} + \partial_b(x). \quad (16.1)$$

Indeed,

$$\binom{A+t}{b+t} - \binom{A}{b} = \sum_{j=1}^t \binom{A+j-1}{b+j}$$

is a diagonal binomial shell above the canonical b -cascade of x . The strict inequality $x < \binom{A}{b}$ is exactly the condition that no carry enters the shell, so termwise shadowing proves (16.1). If that condition ever fails during peeling, the recurrence has already reached the next capacity shell and overflow follows immediately or after the next positive load.

For clarity, write the transfer as an induction. Let x_b denote the upper-core recurrence, so $x_b = a_b$ for $3 \leq b \leq r+2$, and use $x_{r+3} = 0$ at the empty level above the core. Repeated application of (16.1), starting at the top of the full profile, gives

$$M_{b+r} = \binom{B}{b+r} - \binom{A}{b} + x_b \quad (3 \leq b \leq r+3). \quad (16.2a)$$

Each applied step has $b < A$ and uses (16.0); the last identity application is the $b = 4$ step, so no assumption that a_3 fits on A is needed. At $b = 3$,

$$M_{r+3} = \binom{B}{r+3} - \binom{A}{3} + a_3 = \binom{B}{r+3} + e_r \geq \binom{B}{r+3} + 1. \quad (16.2)$$

Define a residual sequence by

$$\nu_{r+3} = 1, \quad \nu_{k-1} = r + \partial_{k-1}(\nu_k) \quad (k = r+3, r+2, \dots, 4). \quad (16.3)$$

The shadow index is $k - 1$ because, after separating the leading shell term $\binom{B}{k}$, the residual ν_k occupies cascade positions at most $k - 1$. Whenever

$$\nu_k < \binom{B}{k-1},$$

the decomposition

$$\binom{B}{k} + \nu_k$$

is a canonical k -cascade, and therefore

$$\partial_k \left(\binom{B}{k} + \nu_k \right) = \binom{B}{k-1} + \partial_{k-1}(\nu_k). \quad (16.3a)$$

This is the exact shell-peeling identity used below: a diagonal shell plus a subcritical residual shadows independently.

Let c_i be the L4 recurrence:

$$c_{r+1} = r, \quad c_i = r + \partial_{i+1}(c_{i+1}).$$

Since

$$\nu_{r+2} = r + \partial_{r+2}(1) = 2r + 2 \geq c_{r+1},$$

monotonicity gives

$$\nu_3 \geq c_2 > \binom{A}{2}. \quad (16.4)$$

Starting from (16.2), peel the shell $\binom{B}{k}$ downward. The no-carry branch is $\nu_k < \binom{B}{k-1}$, exactly the condition that the shell followed by the residual is canonical. If instead, at some level including $k = 3$,

$$\nu_k \geq \binom{B}{k-1},$$

then

$$M_k \geq \binom{B}{k} + \binom{B}{k-1} = \binom{B+1}{k}.$$

A strict inequality is immediate failure; equality is followed by a positive prescribed load and therefore fails one level later. For $k = 3$, this means failure at level 3 or 2.

Otherwise, in particular $\nu_3 < \binom{B}{2}$, the shell-peeling identity (16.3a) applies at every step, and

$$M_3 \geq \binom{B}{3} + \nu_3.$$

Using (16.4),

$$\partial_2(\nu_3) \geq A + 1.$$

Therefore

$$\begin{aligned} M_2 &\geq r + \binom{B}{2} + (A + 1) \\ &= \binom{B+1}{2} + 1. \end{aligned}$$

Thus the full profile at $n = 2r + 5$ is impossible.

17. Conclusion

We first record the missing-level reduction. For $r \geq 11$, levels 0 and n cannot occur in an r -multiplicity antichain. If level 1 occurs, choose r singleton members. Every other member avoids those r points, so only the sizes

$$1, 2, \dots, n - r$$

can occur, giving at most $n - r < n - 3$ occupied levels. Complementation excludes level $n - 1$ from any family with $n - 3$ occupied levels. Hence an $n - 3$ -level witness must occupy exactly

$$2, 3, \dots, n - 2.$$

The asymmetric profile theorem now applies directly: at least r members on these levels force the recurrence capacities, while fitting capacities construct a family with exactly r members on them. No trimming is needed.

For every $r \geq 11$, Section 16 proves failure at $n = 2r + 5$, while Sections 14-15 construct the full profile for every $n \geq 2r + 6$. Consequently

$$n_0(r) = 2r + 5 \quad (r \geq 11).$$

Together with He-Tang's exact values for $r = 2, 3$ and the finite theorem in `SMALL_R_NOTE.md`, this gives the complete threshold classification

$$n_0(r) = \begin{cases} 3, & r = 2, \\ 8, & r = 3, \\ 2r + 4, & 4 \leq r \leq 10, \\ 2r + 5, & r \geq 11. \end{cases}$$

The least constant in He-Tang's bounded-error question $n_0(r) \leq 2r + C$ for all $r \geq 4$ is therefore $C = 5$.

18. Audit and publication status

Four separate adversarial AI review passes dated 2026-07-16 attempted to falsify the lower-window construction, the L4 reverse envelope, the profile-theorem foundations, and the final assembly. None found a mathematical error in a load-bearing argument. The first two audits drove the release-critical repairs already incorporated above. Audit C found only twelve minor or typographical issues; the present revision incorporates all of them, including the two false but non-load-bearing sentences in the technical note and L4 note.

The repository includes the companion L4 proof, finite verifiers, exact $r = 11$ witnesses, a complete Lean development for $r \geq 4$, and explicit statement-correspondence and axiom-audit files.

The result remains AI-assisted and has not undergone independent specialist peer review. Lean verifies the original-formulation least-threshold theorem for every $r \geq 4$; the finite sub-ranges use three disclosed `native_decide` certificates, while the $r \geq 378$ endpoint is purely

symbolic. Specialist human scrutiny is still strongly encouraged, especially for statement correspondence, the finite checker soundness bridges, Sections 5–8, Section 16, and the companion L4 proof.

19. References

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